

Information Gathering: Unobstrusive Methods

CSE 4407

Ajwad Abrar

Junior Lecturer

Department of Computer Science and Engineering
Islamic University of Technology

June 4, 2026



Sampling: The What and Why

- **What:** Systematically selecting *representative* elements of a population
 - **Goal:** Analyze the sample to understand the *whole* population (documents, people, data, etc.)
 - **Analyst's Challenge:** Which documents? Which people?
- **Why**
 - **Costs:** Examining *everything* is expensive! → Time = Money
 - **Speed:** Faster insights by looking at less (but relevant) data
 - **Effectiveness:** Deeper analysis on a smaller set; better chance to follow up on details
 - **Bias:** Avoid skewed perspectives from non-representative individuals or data points

Designing Your Sample: The Blueprint

- **Overview:** To get meaningful results, your sample needs a plan!
- **Steps**
 - **Data:** What *exactly* are you trying to find out or describe? → Identify variables, attributes
 - **Population:** What is the *entire* group your sample should represent? (e.g., last 6 months reports? All managers? Website users?)
 - **Sample Type:** How will you select the elements?
 - **Sample Size:** How many elements do you *need*?

Types of Samples: Picking Your Players

- **Non-Probability:** Selection isn't random
 - **Convenience:** Easiest, but riskiest! (e.g., "Anyone free Tuesday morning?")
 - **Purposive:** Based on judgment (e.g., picking "knowledgeable" people)
- **Probability:** Every element has a known chance of selection
 - **Simple Random**
 - Equal chance for all (like drawing names from a hat)
 - Needs a numbered list
 - **Complex Random**
 - **Systematic:** Every k th item (e.g., every 10th invoice) → Watch out for hidden patterns
 - **Stratified:** Divide population into subgroups first, then sample from each (e.g., sample managers, analysts, programmers separately) → Ensures representation
 - **Cluster:** Select groups assumed to be representative (e.g., pick 2 out of 20 help desks to study)

Sample Size: How Many Are Enough?

- **What:** It's the *absolute number* in the sample that matters most, not just the percentage of the population
 - **Analogy:** Tasting 1 spoonful of soup tells you about the pot, whether it's a small pot or a huge cauldron
 - Sampling 20 people can be effective whether the population is 200 or 2,000,000 (if done right!)
- **Factors Influencing Size**
 - Cost and Time (Practical constraints)
 - Desired Confidence Level (How sure do you want to be?)
 - Acceptable Interval Estimate (How much error can you tolerate?)
- **Two Scenarios**
 - Sampling Quantitative Data (e.g., forms, records) → Use statistical formulae
 - Sampling People (e.g., interviews) → Use guidelines and judgment

Sample Size for Data: The Formula Approach

- **Goal:** Determine n (sample size) needed to estimate a proportion (e.g., % of forms with errors)

- **Steps**

- **Attribute:** What are you measuring? (e.g., 'Order Form Error')
- **Data Source:** Where is this info? (e.g., 'Order Database'/'Past 6 months forms')
- **Estimate p :** Guess the proportion with the attribute (e.g., Estimate 5% or 0.05 have errors) → Use 0.5 if unsure, but this leads to large samples
- **Acceptable Interval i :** How much error is okay? (e.g., ± 0.02)
- **Confidence Level:** How sure? (e.g., 95%) Look up z-value
- **Standard Error:** $\sigma_p = i/z$
- **Sample Size:** $n = \frac{p \times (1-p)}{\sigma_p^2} + 1$

%	z
99	2.58
98	2.33
97	2.17
96	2.05
95	1.96
90	1.65
80	1.28
50	0.67

Example: A. Sembly Company Orders

- **Problem:** Find the percentage of orders with errors
- **Steps**
 - **Attribute:** Orders with errors (name, address, quantity, model #)
 - **Source:** Order forms from the past 6 months
 - **Estimate p :** Analyst estimate ~5% contain errors
 - **Interval i :** Analyst decides ± 0.02 is acceptable
 - **Confidence z :** 1.96
 - **Calculate σ_p :** $i/z = 0.02/1.96 = 0.0102$
 - **Calculate n :** $\frac{p \times (1-p)}{\sigma_p^2} + 1 = \frac{0.05 \times (1-0.05)}{(0.0102)^2} + 1 \approx 458$
- **Conclusion:** Need to sample 458 order forms
- **Note:** Higher confidence or smaller interval would require a *much larger* sample

Sample Size for Interviews: It's About Time!

- **No Magic Formula:** Determining interview sample size is more **art** than science
- **Key Factor**
 - Time commitment (yours and the interviewee's)
 - In-depth interviews are time-consuming
- **Rule of Thumb**
 - Interview **at least 3 people** from *each* organizational level involved (e.g., execs, managers, operational staff)
 - Interview **at least 1 person** from *each* functional area affected (e.g., Sales, Marketing, Production, IT)
- **Stratification:** A well-designed stratified sample means a relatively small number can represent the whole organization effectively
- **Remember:** You do not need huge numbers if you choose wisely → Quality over quantity

Analyzing Quantitative Documents: Playing Detective!

- **What:** The act of discovery and analysis
- **Analyst Role:** Like Sherlock Holmes - seeking clues in hard data
- **Why**
 - Offer insights unavailable elsewhere (historical performance, documented procedures)
 - **Include:** Reports, Records, Forms
 - Show where the organization *has been* and where it *thinks* it's going

Quantitative Docs 1: Performance Reports

- **Purpose:** Compare **Actual** vs. **Intended** Performance
 - Assess the *gap* between goals and reality
 - **Identify trends:** Is the gap widening or narrowing?
- **Examples:** Sales reports (revenue vs. target), Production reports (output vs. quota, defect rates), Service reports (delivery times vs. goals, satisfaction scores vs. benchmarks)
- **Analyst Focus**
 - Is performance measured for key areas?
 - Is the measurement adequate?
 - What story do the trends tell?

Week	Number of Batches Produced	Number of Batches Rejected	Percentage Rejected	Amount Away from 5% Goal
2/2	245	19	7.8	2.8
2/9	229	19	8.3	3.3
2/16	219	14	6.3	1.3
2/23	252	13	5.2	0.2
3/2	245	13	5.3	0.3
3/9	260	13	5.0	***
3/16	275	14	5.1	0.1
3/23	260	13	5.0	***
3/30	260	13	5.0	***
4/6	244	12	4.9	***
4/13	242	11	4.5	***
4/20	249	11	4.4	***
4/27	249	11	4.4	***

*** indicates met or exceeded the <5% goal

Quantitative Docs 2: Records

- **Purpose:** Provide periodic updates on business happenings → Offer rich detail if updated accurately and timely
- **Analyst Focus**
 - **Errors:** Check calculations, totals, data entry mistakes
 - **Design Flaws:** Is the record form confusing? Could it be improved?
 - **Transaction Patterns:** How many? What types? Frequency?
 - **Automation Opportunities:** Can calculation or data manipulation be computerized?

PROJ. NAME										OAK, FC				#		562		KEY SIGNATURE					
RENT POTENTIAL					1175/0		81299				DEPOSIT POTENTIAL								PRORATE			15.121.82	
Base Rent	Refrigerator	Furniture	A/C	Util.	HMSR	T.V.	Maid	Total Rent	Security	Cleaning	31175/0	81299	31700 Tax	Days		Daily Rate		Totals					
855		55						910			H/S dep.	H/S rent		4		30.00/1.20		910.39					
								200		115							Deposits 31.63				340		
PAYMENT RECORD: Tot. 31175/0 + 81299 + Rent = 910												TOTAL INITIAL PAYMENT REQUIRED:										1430.52	
Memo Only		Date Due	Date Paid	Receipt Number	Paid to Noon	Total Rent	Security	Cleaning	31700 Tax	31175/0	81299		Other		Amount Paid	Balance Due							
													Dates	Amt.	Descr.	Amt.							
TV 10/1 MOI		8/10	8/10	106642	8/10	1031.82	202	115	44.88	25					414.82	15	1430.82	0					
		10/1	10/1	107503	10/1	910											910	0					
		11/1	11/1	10935	11/1	485.88											485.88	0					
CIH/S9-16		11/10	11/1	11200	11/10	212.81											212.81	0					
Bill 1 MO		11/10																					
Prorated																							
H/S should be																							
created toward																							
refund deposit																							
Orig. Move-in Date		8-28		d		same		Exp.						X #		1							
BLDG. #						NAME		Kendall						1st									

Quantitative Docs 3: Data Capture Forms

- **Purpose:** Understand the *input* side of the information flow
- **What to Collect**
 - **All forms:** Official and unofficial (“bootleg”), paper and electronic (web forms, fillable PDFs)
 - Blank copies + instructions
 - Examples of filled-in forms (use sampling!)
- **Cataloging Steps**
 - **Collect:** Gather examples of every form
 - **Note Type:** How is it produced/used (printed, handwritten, web, etc.)?
 - **Document Distribution:** Who is supposed to get it?
 - **Compare:** Who actually gets it? (Reality vs. Plan)

Analyzing Data Capture Forms: Official vs. Bootleg

- Key Questions

- **Completeness:** Is it fully filled out? Which fields are skipped? Why? (Poor design? Unnecessary data?)
- **Usage:** Are some official forms ignored? Why? (Bad design? Redundant?)
- **Circulation:** Do forms reach the right people/files? Are online forms accessible? (Workflow/permission issues?)
- **Paper vs. Online:** Compare completion rates if both exists (Usability differences?)
- **Bootleg Forms:** Why do they exist?
 - Simplify a complex official form?
 - Address missing needs?
 - Indicate process problems or workarounds?
 - Sign of political issues? (Control over data input)

Analyzing Qualitative Documents: Reading Between the Lines

- **What:** Written materials revealing expectations, beliefs, values, and human interaction aspects → Emails, Memos, Signs, Web Pages, Manuals, Policy Handbooks
- **Why**
 - Rich source of detail on organizational culture
 - Reveal user expectations and attitudes towards technology
 - Uncover the 'affective, emotional, and motivational aspects' of HCI
- **Scared?** → Learn the systematic approach to decode these documents

Decoding Qualitative Docs: Systematic Guidelines

- **Focus:** Uncovering underlying meanings, values, and relationships
- **Guidelines**
 - **Metaphors:** Look for key or guiding metaphors (e.g., “We’re a family,” “It’s a machine,” “This is war”) → Language shapes behavior
 - **Insiders vs. Outsiders:** Identify an “us vs. them” mentality (e.g., department rivalries, management vs. staff)
 - **“Good” vs “Evil”:** Note terms used repeatedly to describe positive/negative aspects (e.g., “efficient,” “innovative” vs. “bureaucratic,” “legacy”)
 - **Messages/Graphics:** Analyze visuals and text on websites, bulletin boards → What themes are emphasized?
 - **Humor:** Recognize its presence and type → Indicates subcultures, morale, attitudes

Qualitative Docs: Memos and Signs

- Memos

- Analyze content using the 5 guidelines
- Reveals ongoing dialogues values, attitudes, beliefs
- Note sender/receiver patterns (Who talks to whom?) → Often downward/horizontal flow

- Signs and Posters

- Subtle reinforcers of official culture/values
- Examples: “Quality is Job One,” “Safety First,” “Think Customer”
- Give clues about the organization’s stated priorities

Memo

To: All Night Call Desk Staff
From: S. Leep, Night Manager
Date: 2/15/2023
Re: Get Acquainted Party Tonight

It's a pleasure to welcome two new 11-7 Call Desk staff members, Twyla Tine and Al Knight. I'm sure **they'll enjoy working here.** **Being together** in the wee hours makes us feel like **one big happy family.** Remember for your breaks tonight that some of the crew has brought in food. **Help yourself** to the spread you find in the break room, and **welcome to** the clan, Twyla and Al.

Qualitative Docs: Corporate Websites

- **Scope:** Examine B2C and B2B sites
- **Analysis Dimensions**
 - **Content**
 - Apply 5 guidelines
 - Clarity of message?
 - **Design:** Use of color, graphics, animation, layout
 - **Technical:** Interactivity? Accessibility? Security? Customization/Personalization options?
 - **Managerial**
 - Discrepancies between site and stated goals?
 - Does it reflect reality?
- **Relevance**
 - Even if not redesigning the site, it impacts the overall system view
 - Shows the company's external face
 - Ask for usage metrics if relevant

Qualitative Docs: Manuals and Policy Handbooks

- Manuals

- Describe the **ideal** way things *should* be done
- Apply the guidelines to language and tone
- Reality Check
 - Are printed manuals up-to-date?
 - Are they actually used?
 - Compare ideal procedure to actual practice observed elsewhere

- Policy Handbooks

- Cover broad employee/corporate conduct
- Focus: Policies related to IT - computer use, access, security, data handling, service charges
- Reveal guiding corporate values, attitudes, and beliefs regarding technology and information

Using Text Analytics: Making Sense of Unstructured Data

- **Challenge:** Organizations are flooded with Unstructured Qualitative Data
 - **Examples:** Emails, interview transcripts, reports, wikis, blogs, chat logs, social media posts (Facebook, TikTok, $\mathbb{X}$, etc.), customer reviews
- **What:** Software that analyzes this “soft” data to find patterns, themes, and insights
- **Goal:** Understand what customers, employees, and vendors are thinking, feeling, and saying → Discover motivations and values

The Power Of Text Analytics

- **Benefits**
 - Gain valuable insights into customer/vendor/employee perspectives
 - Understand motivations, values, and opinions expressed freely
 - Identify emerging trends or issues mentioned in open text
- **Contrast:** Unstructured vs. Structured Data
 - **Text Analytics:** Analyzes Unstructured Qualitative Data (text, conversations) → Provides qualitative insights
 - **Data Analytics:** Analyzes Structured Quantitative Data → Uses queries/algorithms for quantitative insights
- **Relevance:** Crucial for companies with significant online presence or lots of text-based communication

Under the Hood: Text Analytics Basics

- General Process

- **Input:** User submits documents (transcripts, reports, web data)
- **Preprocessing:** User defines “stop words” (e.g., ‘a’, ‘the’, ‘is’) to ignore
- **Analysis:** Software performs keyword counts, identifies co-occurring words → Concepts
- **Output:** Generates visualizations and summaries

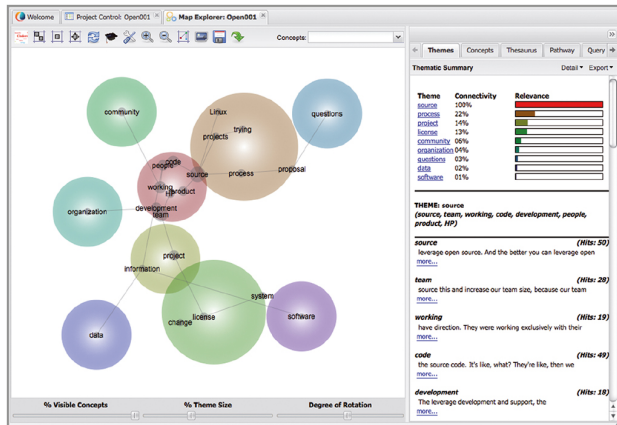
- Leximancer (Example Tool)

- Focuses on rapid, visual, qualitative analysis
- **Identifies Concepts:** Words that frequently appear together, forming themes
- Creates a “thesaurus” of related terms based on context
- Generates visual outputs like concept maps and ranked lists

Making Text Visual: Leximancer Outputs

- Concept Maps

- Show relationships *between* concepts visually.
- Proximity and connecting lines indicate strength of relationship.
- Size of concept 'bubbles' can indicate frequency/importance.



©2024 Pearson Education, Inc.

Making Text Visual: Leximancer Outputs

- Ranked Concept Lists

- Show the most prominent concepts within specific categories or overall.
- Often displayed as bar charts showing frequency or relevance scores.

Category: community				Category: license			
Concept	Rel Freq (%)	Strength (%)	Prominence	Concept	Rel Freq (%)	Strength (%)	Prominence
development	2	5	 4.1	project	11	4	 8.6
information	2	5	 4.1	system	5	3	 7.1
source	4	4	 2.9	source	5	2	 3.7
system	< 1	< 1	 0.0	team	< 1	< 1	 0.0
project	< 1	< 1	 0.0	organization	< 1	< 1	 0.0
team	< 1	< 1	 0.0	development	< 1	< 1	 0.0
organization	< 1	< 1	 0.0	product	< 1	< 1	 0.0
product	< 1	< 1	 0.0	information	< 1	< 1	 0.0

©2024 Pearson Education, Inc.

Text Analytics: More Than Just a Tool

- Your Role
 - **Demonstrate Value:** Show the organization how analyzing unstructured data yields actionable insights and predicts trends
 - **Design Human Processes:** Implementing the tool isn't enough! Design *how* people will use it, interpret results, and integrate insights into decisions
 - **Guide Interpretation:** Help users understand the *qualitative meaning* behind the visuals. It's NOT just about converting text to numbers.
 - **Facilitate Action:** Help translate insights from text analytics into well-reasoned recommendations and organizational improvements
- **Key:** Develop systems that *include* text analytics as a useful means to a qualitative end

Observing Decision Makers: Beyond the Interview

- **What:** Another *unobtrusive* technique for systems analysis.
- **Purpose**
 - Understand what decision makers *actually do*, not just what's documented or said
 - Observe interactions with their physical/ergonomic environment
 - See relationships with other organizational members firsthand
 - Assess Human-Computer Interaction (HCI) - how well systems fit the user

Capturing the Dynamic Nature of Decision Making

- Challenge
 - Managerial work is often fragmented - “a series of interruptions punctuated by short bursts of work”
 - Difficult to capture fully via interviews
- Value: Allows the analyst to see firsthand:
 - How managers gather information
 - How they process it
 - How they share it with others
 - How they use information and technology to accomplish tasks

A Humanistic Approach: The Playscript

- **Concept:** Describe observed actions like a script for a play
- **Structure**
 - **Left Column:** Actor → The decision maker being observed
 - **Right Column:** Script → Their actions, described using action verbs
- **Example Verbs:** Asks, Prints, Discusses, Reads, Compares, Inputs, Observes, Calls, Drafts, Sends, Rewrites
- **Value**
 - Provides an organized, systematic record of observed activities
 - Helps determine information requirements for decisions
 - Reveals dependencies
 - Shows information sources

Demo: **Playscript Example**

Observing the Physical Environment: Clues in the Workspace

- **Concept:** Decision makers influence, and are influenced by, their physical work environment (office)
- **Purpose:** Gain insights into human information requirements by observing the workspace
- **Relationship to Other Methods:** Observing complements interviews, questionnaires by confirming or contradicting the gathered narrative

A Framework for Observation: STROBE

- **Analogy:** Like a 'mise-en-scène' analysis in film criticism
 - Systematically analyzing elements within the frame to understand meaning
 - Can the environment confirm or contract the 'dialogue' (interview/survey/data)?
 - The 'Room Rater' phenomenon!
- **Method:** **STR**uctured **OB**servation of the **E**nvironment
- **Goal:** Systematically examine concrete elements of the office environment

STROBE: Key Observable Elements (1-3)

- Office Location

- Who gets the corner office?
- Are key decision-makers centrally located or dispersed?
- Is it easily accessible?

- Desk Placement

- Face the door? → Power
- Face away? → Openness
- Encourage interaction or create a barrier?

- Stationary Equipment

- Large filing cabinets? → Personal storage
- Minimal storage? → Reliance on digital/shared
- What kind of equipment is present?

STROBE: Key Observable Elements (4-7)

- **Props**
 - Evidence of PC, smartphone, tablet use?
 - Personal items?
 - Tools fo the trade?
- **External Information Sources:** Presence of trade journals, newspapers, books? → Indicates seeking information beyond internal sources
- **Office Lighting and Color**
 - Bright, fluorescent? → Task-oriented
 - Warmer, incandescent? → Casual
 - Color usage - institutional or personalized?
 - Inviting or sterile?
- **Clothing Worn**
 - Formal suits? Business casual? Uniforms?
 - Reflects organizational culture and individual's positioning

Practical Application: STROBE Checklist

- **Method** → Anecdotal Checklist
 - Identify key themes/narratives from interviews/documents
 - Systematically observe the environment using the 7 STROBE elements
 - Compare observations to the narratives
 - Use shorthand symbols on a checklist to record the comparison
- **Goal:** Structured Analysis of how observations confirm, contradict, modify, or supplement the existing organizational narrative

STROBE Checklist Symbols

- **Symbol Key**
 - **Tick:** Confirms the narrative
 - **Cross:** Negates the narrative
 - **Oval/Eye:** Cue to investigate further
 - **Square:** Modifies the narrative
 - **Circle:** Supplements the narrative
- **Example Use:** Checklist compares narratives against observations in different STROBE categories

Demo: **STROBE Example**